Chapter 16: **Generalized Colorings of Planar Graphs**

We saw that the fundamental result about coloring planar graphs is that the chromatic number is at most 4. Further, every planar graph has at most $3n - 6$ edges, where n is the order, and thus a vertex of degree at most 5. We consider here variations: replace planar graph by something more general or less general, and/or replace chromatic number by something stronger or weaker.

16.1 Extensions on Planarity

A graph is 1-***planar*** if it can be drawn such that every edge intersects at most one other edge.

THEOREM. *(Borodin) The chromatic number of a* 1*-planar graph is at most* 6*.*

A graph is k-***thick*** if its edges can be partitioned into k planar graphs. Since a planar graph has average degree less than 6, a k-thick graph has average degree less than $6k$. In particular, a 2-thick graph has a vertex of degree at most 11, and hence (by induction) chromatic number at most 12. It is not known if this is best possible:

QUESTION. *What is the maximum chromatic number of a* 2*-thick graph?*

This is sometimes called the earth-and-moon problem.

16.2 Restrictions on Planarity

A graph is ***outerplanar*** if if can be drawn in the plane such that all vertices are on the outer cycle. A maximal outerplanar graph has exactly $2n-4$ edges, and is formed by taking a hamilton cycle and triangulating the interior. An outerplanar graph can be characterized as having neither subdivision of K_4 nor a subdivision of $K_{2,3}$ as a subgraph.

THEOREM. *The chromatic number of an outerplanar graph is at most* 3*.*

In fact, the key to this is that every outerplanar graph has a vertex of degree at most 2. (Proof: induction.) We say that a graph is k-***dependent*** if it and every subgraph has a vertex of degree at most k. So a tree is 1-dependent, an outerplanar graph is 2-dependent, and a planar graph is 5-dependent.

Because of the characterization of Eulerian graphs, the term ***Eulerian*** is often used to mean every degree is even:

THEOREM. *The chromatic number of an Eulerian maximal planar graph is at most* 3.

(Proof: nice! Induction.) In fact there is a converse: if a planar graph is 3-colorable then it can be extended to an Eulerian planar graph.

A harder theorem is:

THEOREM. *(Grötzsch) A triangle-free planar graph is* 3*-colorable.*

A nice 5-color theorem is:

THEOREM. *(Borodin) A planar graph has a proper* 5*-coloring such that there is no* 2*-colored cycle.*

16.3 Two-Color Theorems

We noted that outerplanar graphs are 3-colorable. So what about coloring them with 2 colors?

It is not hard to show that one can partition the vertices of an outerplanar graph into two forests. (The minimum number of forests that one can partition the vertex set into is called the graph's (vertex) ***arboricity***.) But one can improve that:

THEOREM. *One can partition the vertices of an outerplanar graph into an independent set and a forest.*

PROOF. Induction. Remove vertex v of minimum degree, apply the inductive hypothesis, and then v can be added either to the independent set or to the forest. (Actually true for any 2-degenerate graph.) ◻

A ***linear forest*** is a graph that is the vertex-disjoint union of paths. Several authors showed the following result. But there is a simpler proof.

THEOREM. *One can partition the vertices of an outerplanar graph into two linear forests.*

PROOF. Start at any vertex v. Let A be the vertices at even distance from v and B the vertices at odd distance from v. We claim that this is a suitable partition.

Note: we only need to show that the vertices at a specific distance from v form a linear forest. Let L be such a layer. Can check that a $K_{1,3}$ in L would mean a subdivided $K_{2,3}$ in the graph, which is not outerplanar. Also a cycle in L would mean a subdivided K_4 in the graph, which is not outerplanar. ⌑

(Woodall extended these ideas, and looked at choosability.)

THEOREM. *One can partition the vertices of an planar graph into two outerplanar graphs.*

PROOF. Again partition the vertices by parity of distance from some vertex. ⌑

Note that there is a similar, but much harder, result for edge partitions.

16.4 Three-Coloring a Planar Graph

It is not hard to show that one can partition the vertices of a planar graph into three forests. (Induction.) But actually Stein proved:

THEOREM. *One can partition the vertices of an planar graph into two forests and an independent set.*

This was then strengthened by Borodin's 5-color theorem about acyclic coloring.

Cowen, Cowen, and Woodall showed that one can partition the vertices of a planar graph into three graphs of maximum degree at most two. This was extended independently by Poh and the author. The proof uses strong induction.

Theorem. *One can partition the vertices of a planar graph into three linear forests.*

Given a coloring of the vertices, we define a ***color component*** as a component of the subgraph induced by any of the colors. So the statement of the theorem is equivalent to saying there is a 3-coloring where every color component is a path.

Proof. We prove a stronger statement by induction: that if any two adjacent vertices u and v are designated, it is possible to color the graph G into three linear forests such that neither u nor v has a neighbor of their own color.

The base case is trivial; e.g. a graph with say two vertices. In general, if say vertex u has degree at most 2, we can remove it, apply the inductive hypothesis to what remains with designated vertices v and any of its neighbors, and then add back u giving it the color missing from its neighborhood. So we may assume both u and v have degree at least 3.

Proceed as follows. Subdivide the edge uv by adding a vertex x in the middle of the edge. Then add edges to make the graph maximal planar without re-adding the edge uv. Since this is a maximal planar graph, the neighborhood of vertex u contains a cycle, and such a cycle separates u from v. Out of all separating cycles, let C be one that is as short as possible. Note that C contains x. Further C does not have any chords, since otherwise one obtains a shorter separating cycle.

We can think of C drawn such that u is inside and v is outside. Let G_u be the graph obtained if one removes all vertices outside C and contracts all of C to a single vertex, call it c. Note that G_u is planar, and that vertices u and c are adjacent. So one can apply the inductive hypothesis to G_u to obtain a suitable 3-coloring where neither u nor c has a neighbor of their color. Similarly, let G_v be the graph obtained if one removes all vertices inside C and contracts all of C to a single vertex, called c. Apply induction hypothesis to it.

We can assume that in G_u vertex u gets color 1 and vertex c gets color 2, and in G_v vertex v gets color 3 and vertex c gets color 2. Then take this coloring and lift it back to G so that all of C gets color 2. Note that in this coloring of G vertices u and v have no neighbor of their color. Further, since c had no neighbor of its color, no vertex of color 2 is adjacent to C. And since x is not a real vertex, in G we have only the color component $C - x$, which is a path. ⌑

16.5 Colorings of Sparse Planar Graphs

There are many results on some generalized version of coloring on planar graphs with large giirth. A common technique is to show that planar graphs with high girth are sparse, and that the coloring result is is true for all sparse graphs. The former uses Euler's formula, like we've done before:

FACT. *If a planar graph has girth at least g, then the average degree is less than $2g/(g-2)$.*

PROOF. Recall that $V - E + F = 2$. Further, the boundary of every face must have length at least g, but as before each edge is on the boundary of (at most) two faces, and thus we have $2E \geq gF$. Putting these two formulas together, some arithmetic yields $2E/V < 2g/(g-2)$. ◻

The ***maximum average degree*** of a graph G, denoted $mad(G)$, is the maximum over all induced subgraphs of the average degree. Since subgraphs of planar graphs are planar, the above fact also holds for maximum average degree.

There is a general strategy to prove coloring results for sparse graphs. The first step is to identify some situations that allow induction. The second step is to show that when none of these situations occurs, there is a lower bound on the average degree. The latter is often proven by ***discharging***. We give an example. Note that better results are known.

THEOREM. *If a graph G has $mad(G) < 5/2$, then one can 2-color the vertices such that every vertex has at most one neighbor of its color.*

PROOF. Three situations allow induction: (a) a vertex of degree at most 1; (b) two adjacent vertices of degree 2; (c) a degree-3 vertex all of whose neighbors are degree 2. In each case one removes the one/two/four vertices, applies the inductive hypothesis to the remaining graph H, gives each deleted vertex the color opposite to its neighbor in H, and in (c), gives the degree-3 vertex the minority color from its neighborhood.

The final step is to show that if a graph has none of the three situations described above, then its average degree is at least 5/2. Start by giving each vertex a "charge" equal to its degree. At the beginning, the average charge equals the average degree. Then the ***discharging rule*** is that each vertex of degree at least 3 sends $\frac{1}{4}$ charge to each degree-2 neighbor.

Every degree-2 vertex receives charge from both its neighbors, and so it ends up with charge exactly $5/2$. Every degree-3 vertex sends charge to at most two neighbors, and so it ends up with charge at least $5/2$. Every vertex with larger degree ends with at least $\frac{3}{4}$ of its original charge, and so ends up with charge at least 3. In other words, at the end every charge and hence the average charge is at least $5/2$. But this equals the average charge at the beginning. ⌑